

MODEL SELECTION AND EVALUATION REPORT

BICT242: Data Scalability and Analytics

Student Course Recommendation System

1. Model Selection

A content-based filtering approach was selected for this prototype. Content-based filtering recommends items that are similar to the characteristics of items or content that match a user's stated interests. This approach was selected because the prototype has course descriptions and student interests but does not have a large historical database of real users and ratings.

2. TF-IDF

TF-IDF stands for Term Frequency-Inverse Document Frequency. It converts text into numerical weights. Words that are useful for distinguishing one course from another receive greater importance than words that occur very frequently across the dataset.

3. Cosine Similarity

After TF-IDF conversion, cosine similarity measures the similarity between the student's interest vector and each course vector. A higher similarity score means the course text is more closely related to the student's interests.

4. Recommendation Process

The student enters interests such as Python, data analysis or programming.

The interests are converted into a TF-IDF vector.

The system compares this vector with all course vectors.

Cosine similarity produces a score for every course.

A small bonus is added when the selected difficulty level matches the course level.

The courses with the highest scores are returned as recommendations.

5. Evaluation Method

The prototype was evaluated using 20 simulated student profiles. Each profile has one target course that represents a relevant course for that student. The model generates its top three recommendations. A hit occurs when the target course appears in those three recommendations.

6. Evaluation Metrics

Metric	Result	Interpretation
Precision@3	0.333	Proportion of the three recommended positions that contained the target course, averaged using the single-target setup.
Recall@3	1.000	Proportion of target courses that appeared within the top three recommendations.
F1@3	0.500	Harmonic mean of Precision@3 and Recall@3 for this evaluation setup.
Hit Rate@3	1.000	Percentage of student profiles for which the target course appeared in the top three.

7. Results

Using the synthetic evaluation dataset, the system achieved a Precision@3 of 0.333, Recall@3 of 1.000, F1@3 of 0.500, and a Hit Rate@3 of 1.000. These results demonstrate how the prototype can be evaluated using measurable recommendation metrics.

8. Example Recommendations

Student interests: Python programming data | Level: Beginner

Python Programming (Programming) similarity 0.718

Data Analytics (Data Science) similarity 0.421

Java Programming (Programming) similarity 0.380

Student interests: IoT sensors microcontrollers | Level: Intermediate

Internet of Things (IoT) similarity 0.648

Embedded Systems (IoT) similarity 0.414

Cybersecurity (Security) similarity 0.077

Student interests: cybersecurity networks encryption | Level: Intermediate

Cybersecurity (Security) similarity 0.533

Computer Networks (Networking) similarity 0.217

Internet of Things (IoT) similarity 0.074

Student interests: cloud deployment automation | Level: Advanced

DevOps (Cloud) similarity 0.594

Cloud Computing (Cloud) similarity 0.563

Embedded Systems (IoT) similarity 0.096

9. Strengths

Simple to understand and implement.

Does not require a large historical user-rating database.

Recommendations can be generated immediately from course descriptions.

The approach can be extended with ratings, clicks and enrolment history.

10. Limitations and Future Improvements

Use a larger real student dataset if available.

Add student ratings and course enrolment history.

Develop collaborative filtering and compare it with content-based filtering.

Create a hybrid model that combines content and collaborative information.

Add more detailed student features such as completed modules and career interests.

Monitor recommendations over time and retrain the model when new data becomes available.

11. Conclusion

The selected content-based model provides a practical recommendation approach for the educational domain. TF-IDF and cosine similarity allow the system to compare student interests with course descriptions and return ranked recommendations. The evaluation demonstrates a repeatable way to measure the prototype, while the limitations identify clear areas for future development.